Forecasting Daily Direction of U.S. Wheat Futures Using Machine Learning Models and Investment Strategies

Active Portfolio Management with Machine Learning and Time Series Forecasting

New York University

Mathew Martin

mm14460@nyu.edu

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Abstract

This report summarises the work completed throughout the VIP course on ML for Agricultural Commodity Markets. The core task is binary classification of the next-day price direction (Up / Down) of U.S. Wheat Futures (ticker: ZW=F), using a combination of lagged price features and FRED-MD macroeconomic indicators. Four machine learning models are implemented: a Decision Tree, a Vector Autoregression (VAR), a Gated Recurrent Unit (GRU), and a Transformer encoder. Two investment strategies — Long-Only Momentum and Long-Short Momentum — are then constructed on top of these forecasting signals. The report discusses the data pipeline, model design choices, evaluation results, and the translation of model predictions into trading performance. A key finding is that VAR, despite its simplicity, outperforms the deep learning models on this limited dataset, while both neural models suffer from class collapse due to class imbalance. The strategies reveal that near-random signals require careful risk management, and that predictive accuracy does not straightforwardly translate into positive economic value.

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1 Introduction and Motivation

U.S. Wheat Futures (ZW=F) are one of the most actively traded agricultural commodity contracts globally, serving as a benchmark for global food supply expectations, geopolitical risk, and macroeconomic conditions. Accurate forecasting of wheat price movements carries significant practical value for commodity traders, portfolio managers, and agribusiness hedgers seeking to manage exposure to agricultural price risk.

Rather than forecasting price *levels*, this project focuses on predicting the daily *direction* of price change — whether the closing price on day t will be higher or lower than on day $t-1$. This formulation is particularly relevant for discretionary and systematic trading, where the primary decision is whether to hold a long or short position, and where errors in predicting the sign of returns can be more costly than errors in predicting their magnitude. Direction prediction also maps naturally onto binary classification, enabling the use of a broad range of machine learning methods.

Macroeconomic conditions play an important role in commodity price dynamics. Interest rates affect the financing cost of commodity inventories and the opportunity cost of holding physical stocks. Credit spreads signal broader risk appetite in financial markets that often spills over into commodity flows. Exchange rates shift the relative purchasing power of international buyers. Consumer sentiment and industrial activity reflect demand-side conditions for energy and food inputs. Incorporating FRED-MD macroeconomic indicators alongside price history is therefore well-motivated: the macro environment provides context that pure technical models based only on past prices may miss, especially during regime changes such as monetary policy tightening cycles.

This report covers the following work:

- **Four forecasting models**: Decision Tree, VAR(3), GRU, and Transformer encoder, all implementing a shared `BaseForecastModel` abstract class.
- **Two investment strategies**: Long-Only Momentum and Long-Short Momentum, constructed from VAR and Transformer signals and evaluated against a buy-and-hold benchmark.

The goal is not to identify a single "best" model, but to understand the trade-offs between different modelling approaches — statistical versus deep learning, parsimonious versus over-parameterised — and to explore whether any forecasting edge translates into real economic value.

2 Data and Preprocessing

2.1 Target Variable Construction

The binary target variable y_t is defined as:

$$y_t = \mathbf{1}\left[\, P_t > P_{t-1} \,\right]$$

where P_t denotes the daily closing price of ZW=F on day t. A value of $y_t = 1$ indicates an "Up" day; $y_t = 0$ indicates a "Down" day. By construction, the target at time t is determined only by prices up to and including day t; the features used to predict it are constructed entirely from information available at the close of day $t-1$ or earlier, ensuring strict no-look-ahead compliance.

2.2 Price Features: Lagged Closing Prices

Thirty lagged closing prices $\{P_{t-1}, P_{t-2}, \ldots, P_{t-30}\}$ are included as features, forming a 30-day lookback window. This rolling window structure means that at each time step t, the model receives only past prices and never any contemporaneous information about day t. After each prediction, the window is slid forward by one day.

2.3 FRED-MD Macroeconomic Indicators

The macroeconomic features are drawn from the FRED-MD dataset, a large monthly database of U.S. macroeconomic time series maintained by the Federal Reserve Bank of St. Louis. A curated set of 31 variables was selected, covering:

- **Real activity**: Real Personal Income (RPI), Real Retail and Food Services Sales (CMRMTSPLx), Industrial Production indices.
- **Interest rates and yield curve**: Federal Funds Rate (FEDFUNDS), 3-month and 6-month T-bill rates (TB3MS, TB6MS), 1-, 5-, and 10-year Treasury yields (GS1, GS5, GS10).
- **Credit spreads**: Moody's AAA and BAA corporate bond yields; spread variables (AAAFFM, BAAFFM) indicating risk appetite.
- **Exchange rates**: USD/CHF, USD/JPY, GBP/USD, USD/CAD — relevant because wheat is traded globally in USD.
- **Prices**: Producer Price Index for commodities (PPICMM).
- **Sentiment**: University of Michigan Consumer Sentiment (UMCSENTx).
- **Finance**: S&P 500 index, S&P P/E ratio, consumer credit.

2.4 T-Code Stationarity Transformations

FRED-MD provides a "t-code" for each variable specifying the recommended stationarity transformation. Applying these transformations is essential because most macroeconomic series are non-stationary in levels; regression models applied to non-stationary inputs produce spurious results. The six transformations are:

T-code	Transformation
1	Level (no transformation)
2	First difference: Δx_t
3	Second difference: $\Delta^2 x_t$
4	Logarithm: $\ln(x_t)$
5	Log first difference: $\Delta \ln(x_t)$ (approximate growth rate)
6	Log second difference: $\Delta^2 \ln(x_t)$

For example, FEDFUNDS has t-code 2 (first difference), transforming the level of the Fed Funds Rate into its month-over-month change. PPICMM has t-code 6 (log second difference), removing both a trend and a drift in growth rates.

2.5 One-Month Reporting Lag

FRED-MD is released with an approximate one-month delay: data for month M typically becomes available in month $M + 1$. To prevent look-ahead bias, all monthly macro data is shifted forward by one period (`shift(1)` in pandas). This ensures that on any trading day t in month M, only macro data up to month $M - 2$ or earlier is used as input. After applying this lag, monthly values are forward-filled onto the daily trading calendar to align frequencies.

2.6 Data Splits and Cross-Validation

The dataset is split chronologically into three non-overlapping periods. After applying the 30-day rolling window and dropping NaN rows, the full feature matrix spans 6,224 trading days (August 2000 – November 2025) with 61 features and a class balance of 47.6% Up-days overall.

Split	Start	End	Size
Training	28 Aug 2000	29 Aug 2014	3,502 days (47.9% Up)
Validation	5 Jan 2015	29 Nov 2024	2,492 days (47.5% Up)
Test	3 Jan 2025	28 Nov 2025	228 days (46.1% Up)

Shuffling is strictly avoided: financial time series have temporal dependencies, and shuffling would allow future information to leak into the training set, invalidating any estimate of

out-of-sample performance.

For hyperparameter selection, 5-fold `TimeSeriesSplit` cross-validation is used. Unlike standard k-fold, `TimeSeriesSplit` always trains on past data and validates on future data, preserving temporal ordering within each fold. This is critical for time series because a model trained on randomly sampled folds would see future observations during training, leading to optimistic and misleading performance estimates.

2.7 Scaling and Leakage Prevention

`StandardScaler` is fitted *exclusively on the training set* and then applied to the validation and test sets without refitting. Fitting the scaler on the full dataset before splitting constitutes data leakage: the mean and variance computed over future observations would encode information that the model could not have access to in a live deployment setting. Similarly, within each cross-validation fold, the scaler is re-fitted on the training portion of that fold only.

3 Forecasting Models

All four models implement the shared `BaseForecastModel` abstract base class with five required methods: `fit`, `predict`, `evaluate`, `save`, and `load`. The `evaluate` method returns MSE, RMSE, MAE, and R^2 for regression formulations, and accuracy, precision, recall, F1-score, and ROC-AUC for classification formulations.

3.1 Decision Tree

Conceptual background. A Decision Tree is a non-parametric supervised learning algorithm that partitions the feature space into rectangular regions by recursively applying binary split rules. Each internal node splits on a single feature threshold; each leaf node assigns a class label. Decision Trees belong to the family of tree-based models and make no distributional assumptions about the inputs. They are highly interpretable but prone to overfitting on high-dimensional or noisy data without regularisation.

Implementation. The tree was implemented using `scikit-learn`'s `DecisionTreeClassifier` with the following hyperparameters selected via `TimeSeriesSplit` cross-validation: `max_depth=5`, `min_samples_leaf=20`, `criterion='gini'`. The 30 lagged prices and 31 FRED-MD macro features (61 total) are used as inputs. Class weights are set to `balanced` to partially correct for the slight class imbalance in daily direction data.

Strengths and limitations. Decision Trees are interpretable and handle non-linear interactions without feature engineering. However, they are sensitive to small perturbations in the training data (high variance), and the recursive partitioning of a 61-dimensional

feature space with limited training samples leads to shallow, poorly generalising splits. Daily price direction in efficient markets is inherently near-random, which further limits tree depth from adding meaningful signal.

Results. The Decision Tree achieved test set accuracy of approximately **50%** and ROC-AUC of approximately **0.50**, consistent with near-random performance. This serves as an important baseline: it confirms that daily wheat futures direction is close to a 50/50 coin flip on the test period, in line with the Efficient Market Hypothesis for liquid futures markets. The confusion matrix showed a slight bias toward predicting the majority class. Performance below 50% on some cross-validation folds reflects the noise sensitivity of tree splits on a nearly balanced binary target.

3.2 Vector Autoregression (VAR)

Conceptual background. Vector Autoregression (VAR) is a multivariate time series model in which each variable is expressed as a linear function of its own past values and the past values of all other variables in the system. A VAR of order p (VAR(p)) is:

$$\mathbf{y}_t = \mathbf{c} + \sum_{i=1}^{p} \mathbf{A}_i \mathbf{y}_{t-i} + \boldsymbol{\varepsilon}_t$$

where $\mathbf{y}_t$ is the vector of endogenous variables, $\mathbf{A}_i$ are coefficient matrices, $\mathbf{c}$ is a constant vector, and $\boldsymbol{\varepsilon}_t$ is a white noise vector. VAR belongs to the family of classical econometric models and assumes stationarity of all included series. Its main assumption is that linear dependencies in the joint history of the system summarise all predictable variation.

Implementation. The VAR was implemented from scratch using NumPy and SciPy, without `statsmodels`, to meet the course requirement of dependency-minimal implementations. The lag order $p = 3$ was selected by AIC. The system includes five endogenous variables: the lagged wheat futures price and four macroeconomic series — FEDFUNDS, GS10, BAA, and TB3MS. These were chosen for their economic relevance to commodity markets:

- **FEDFUNDS**: The Federal Funds Rate directly affects commodity financing costs and speculative flows into futures markets.
- **GS10**: The 10-year Treasury yield captures long-run inflation expectations and real growth expectations, both correlated with commodity demand.
- **BAA**: Moody's BAA corporate bond yield reflects credit risk and risk appetite; credit stress often spills into commodity market de-risking.
- **TB3MS**: The 3-month T-bill rate, together with GS10, captures yield curve shape (slope = economic outlook indicator).

The model generates one-step-ahead forecasts of the wheat price variable, and the binary direction signal is derived by comparing the forecast to the current price: $\hat{y}_t = \mathbf{1}[\hat{P}_t > P_{t-1}]$.

Strengths and limitations. VAR is parsimonious: with $p = 3$ lags and 5 variables, it estimates $5 \times 5 \times 3 + 5 = 80$ parameters. Given 3,502 training observations, this is well-identified. VAR also naturally captures lead-lag relationships between the macro variables and wheat prices. Its main limitation is the linearity assumption: commodity markets exhibit regime-switching behaviour and non-linear dynamics that a VAR cannot capture. It is also sensitive to variable selection — INDPRO and CPIAUCSL, which are theoretically well-motivated for wheat forecasting, were not available in the 31-feature FRED-MD selection used in this project.

Results. Two versions of the VAR were evaluated across the project. The midterm submission (`var-v2.ipynb`) used `statsmodels` with 5 macro variables selected by position (the first 5 columns of the feature matrix: RPI, W875RX1, CMRMTSPLx, IPFPNSS, USWTRADE) and AIC-selected lag order $p = 1$, achieving test accuracy of **53.07%** and ROC-AUC of **0.5438**. For the investment strategy notebooks, the variable selection was corrected to use economically motivated predictors — FEDFUNDS, GS10, BAA, and TB3MS — with fixed lag order $p = 3$, yielding test accuracy of **47.37%** and ROC-AUC of **0.5283**. The drop in accuracy when switching to principled variable selection is itself instructive: the original result was partly driven by spurious correlation between arbitrary macro columns and wheat direction rather than genuine economic signal. The corrected model's above-random AUC of 0.5283 reflects a more honest estimate of predictive content. 5-fold cross-validation accuracy on the training set averaged 0.4923 ($\pm$0.013), confirming modest but consistent signal.

3.3 Gated Recurrent Unit (GRU)

Conceptual background. The Gated Recurrent Unit (GRU) is a recurrent neural network architecture designed to model sequential dependencies while mitigating the vanishing gradient problem of vanilla RNNs. It uses two gating mechanisms: an *update gate* $\mathbf{z}_t$ controlling how much past information is carried forward, and a *reset gate* $\mathbf{r}_t$ controlling how much past information is discarded when forming the candidate hidden state:

$$
\begin{aligned}
\mathbf{z}_t &= \sigma(\mathbf{W}_z \mathbf{x}_t + \mathbf{U}_z \mathbf{h}_{t-1}) \\
\mathbf{r}_t &= \sigma(\mathbf{W}_r \mathbf{x}_t + \mathbf{U}_r \mathbf{h}_{t-1}) \\
\tilde{\mathbf{h}}_t &= \tanh(\mathbf{W}_h \mathbf{x}_t + \mathbf{U}_h (\mathbf{r}_t \odot \mathbf{h}_{t-1})) \\
\mathbf{h}_t &= (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t
\end{aligned}
$$

GRUs belong to the deep learning family of models and make no explicit distributional assumptions, learning complex non-linear temporal patterns directly from data.

Implementation. The GRU was implemented in PyTorch with a single hidden layer of 64 units, a sequence length of 30 (matching the price lookback window), and a final linear layer followed by sigmoid activation for binary classification. The loss function is `BCEWithLogitsLoss`. Training used the Adam optimiser for 50 epochs with a learning rate of 10^{-3} and batch size of 32. The model receives the full 61-feature input at each time step.

Strengths and limitations. GRUs can model non-linear temporal dependencies and long-range patterns that linear models cannot capture. They are parameter-efficient compared to LSTM. However, with approximately 161–185 effective training sequences (after applying the rolling window) and a model with thousands of parameters, the GRU is severely over-parameterised relative to the available data. This leads to poor generalisation. Additionally, the absence of class weighting in the loss function causes the model to predict the majority class on nearly all test observations.

Results. The GRU exhibited **class collapse** on the test set, predicting "Down" (the majority class) for almost every observation. This resulted in test accuracy approximately equal to the majority-class frequency ($\approx 50\%$) and near-zero recall for the minority class, with a precision-recall trade-off heavily skewed toward "Down" predictions. The root cause is the combination of class imbalance in daily direction data and the absence of `pos_weight` in the binary cross-entropy loss. Adding `pos_weight` to upweight the minority class would be the primary corrective step. This failure is itself an important finding: deep learning models need far more data relative to parameter count than is available in this setting, and class imbalance can silently cause complete signal loss.

3.4 Transformer Encoder

Conceptual background. The Transformer architecture replaces the sequential hidden-state computation of RNNs with a self-attention mechanism that directly computes pairwise relationships between all time steps in the input sequence. For binary classification, an encoder-only Transformer is appropriate: the sequence of input feature vectors is passed through multi-head self-attention layers, and the output at the final time step is used for classification. The self-attention operation is:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}$$

where $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ are query, key, and value matrices derived from linear projections of the input.

Implementation. The Transformer encoder used in the investment strategies was configured with sequence length 20, model dimension $d_{\text{model}} = 64$, 4 attention heads, and 2 encoder layers. A positional encoding layer is added to inject sequence-order information (which self-attention alone does not capture). The model was trained with binary cross-entropy loss and Adam optimiser. Saved weights from the midterm submission are loaded for the strategy backtests, avoiding re-training.

Strengths and limitations. Transformers are state-of-the-art for sequence modelling in NLP and have shown strong results for long-horizon financial forecasting when large datasets are available. The self-attention mechanism can identify which historical time steps are most relevant for a given prediction, offering implicit feature importance. However, Transformers are highly over-parameterised for datasets of this size. Like the GRU, the absence of class weighting causes class collapse, and the quadratic memory complexity of self-attention (in the sequence length) is unnecessary for a 20-step input.

Results. The Transformer exhibited partial class collapse on the test set, predicting "Up" on only **41 of 228 test days** (18%) and "Down" on the remaining 187 (82%), yielding test accuracy of **52.63%** and ROC-AUC of **0.4692**. The accuracy figure is misleading: because 53.9% of test days are Down days, a model heavily biased toward Down automatically exceeds 50% accuracy without genuine predictive skill. The sub-0.5 AUC confirms the model's probability scores are negatively correlated with true direction. In the strategy backtests this manifests as near-constant flat positioning (Long-Only, only 4 trades) or near-constant short positioning (Long-Short, only 5 trades), making the Transformer a useful stress-test for how a biased classifier translates directly into poor trading outcomes.

3.5 Model Comparison

Table 1 summarises the key performance metrics and characteristics of all four models.

Table 1: Summary comparison of the forecasting models on the test set (228 days, Jan–Nov 2025).

Model	Test Acc.	ROC-AUC	Key Strength	Key Limitation
Decision Tree	$\approx 50\%$	≈ 0.50	Interpretable baseline	High variance, overfits
VAR (midterm)	53.07%	0.5438	Parsimonious, AIC lag order	Arbitrary variable selec
VAR (final)	47.37%	0.5283	Economically motivated vars	Linear only; best AUC
GRU	$\approx 50\%$	≈ 0.50	Non-linear temporal patterns	Class collapse; over-par
Transformer	52.63%	0.4692	Global attention over sequence	Class collapse; AUC be

The results illustrate a clear trade-off: VAR, despite being the most statistically traditional

model, outperforms the deep learning models on this dataset precisely because it is parsimonious. With approximately 161–185 effective training sequences after applying the rolling window, GRU and Transformer cannot learn generalisable patterns. They have vastly more parameters than their training signal can constrain, leading to class collapse. VAR's 80 parameters are comfortably identified by the 3,502 training observations.

This finding aligns with the broader literature on small-sample financial forecasting: model parsimony often dominates representational capacity when data is scarce. The Decision Tree occupies a middle ground: it is non-linear and can model interactions, but the high-dimensional feature space (61 features) and near-random target make it impossible for shallow trees to find stable partitions. All models perform near chance, consistent with the Efficient Market Hypothesis for liquid futures markets, and near-random classification should be treated as a legitimate and expected outcome rather than a model failure.

4 Investment Strategies

Two investment strategies were implemented on top of the VAR and Transformer forecasting signals. Both strategies are implemented as subclasses of `BaseForecastModel` with the full five-method interface. The backtesting infrastructure is a lightweight, custom-built `Backtester` class using NumPy and pandas, computing daily returns, equity curves, and the three key performance metrics: cumulative return, Sharpe ratio, and maximum drawdown. All strategy signals are shifted by one trading day before computing returns, ensuring that the prediction made on day t is executed at the open of day $t + 1$ — no look-ahead bias in the backtest.

4.1 Strategy 1: Long-Only Momentum

Definition and rules. The Long-Only Momentum strategy enters a full long position (+1) when the forecasting model predicts an "Up" day, and holds a flat position (0) when the model predicts "Down." Position sizing is binary: either fully invested or fully out of the market. Rebalancing occurs daily at the open of the following trading day.

Signal	Position	Interpretation
Predicted Up ($\hat{y}_t = 1$)	Long +1	Buy and hold wheat futures for one day
Predicted Down ($\hat{y}_t = 0$)	Flat 0	Exit or stay out of the market

Economic intuition. Long-Only Momentum is grounded in the momentum literature. Jegadeesh and Titman [1993] demonstrate that assets that have recently outperformed continue to do so over horizons of 3–12 months. Adapted to a daily classification framework, momentum implies: if the model forecasts upward price movement, a rational strategy is to participate in that move by going long. Crucially, flat positions during predicted down-

days avoid losses that a passive buy-and-hold investor would incur, providing downside protection without requiring short-selling infrastructure. Moskowitz et al. [2012] extend this logic to time-series momentum, showing that the sign of past returns predicts future returns across asset classes including commodities.

The strategy was run separately using VAR signals (primary model) and Transformer signals (comparison model), both evaluated against a buy-and-hold benchmark.

Signal model: VAR. The VAR model generates a binary directional forecast by comparing its one-step-ahead price forecast to the current price: $\hat{y}_t = \mathbf{1}[\hat{P}_t^{\text{VAR}} > P_{t-1}]$. During the test period (3 January – 28 November 2025, 228 days), VAR generated 123 trades — a roughly balanced mix of "Up" and "Down" signals — reflecting its modest but real predictive content.

Signal model: Transformer. The Transformer's class collapse resulted in heavily Down-biased signals during the test period. With only 4 trades over 228 test days, the strategy was almost entirely out of the market, producing a cumulative return of −5.54% — marginally worse than buy-and-hold (−4.26%) while missing nearly all up-days. The signal distribution confirms this: 187 of 228 test-period signals were "Down" (flat), with only 41 "Up" signals triggering a long position.

Performance metrics.

Table 2: Long-Only Strategy — backtest results, test period (3 Jan – 28 Nov 2025, 228 days).

Metric	VAR Long-Only	Transformer Long-Only	Buy & Hold
Cumulative Return	−5.37%	−5.54%	−4.26%
Annual Return	−5.94%	−6.13%	−4.72%
Sharpe Ratio	−0.250	−0.495	−0.082
Max Drawdown	−16.40%	−10.82%	−19.49%
Num Trades	123	4	1
Win Rate	47.29%	46.34%	46.02%

Advantages of Long-Only Momentum.

- **Downside protection**: exiting the market on predicted down-days reduces drawdown compared to buy-and-hold, even with a near-random signal, because roughly half the flat days would have been losses.
- **Implementable**: wheat futures are liquid and require no short- selling margin; only long positions are needed.

- **Macro-conditioned signal**: VAR uses FEDFUNDS, GS10, BAA, and TB3MS — economically motivated predictors with documented relationships to commodity financing costs and risk appetite.
- **Transparent signal**: the VAR forecast is fully interpretable; each coefficient can be examined.

Limitations of Long-Only Momentum.

- **Opportunity cost**: false "Down" predictions keep the strategy flat during genuine up-moves, missing gains.
- **Near-random signal**: the corrected VAR's 47.37% test accuracy (AUC 0.5283) provides only a thin edge that is easily eroded by transaction costs not modelled here.
- **No position sizing**: the strategy commits a full unit regardless of signal confidence; a Kelly-scaled or volatility-scaled variant would be more risk-aware.
- **Transformer inapplicability**: the Transformer's class collapse makes its Long-Only contribution trivially small.

4.2 Strategy 2: Long-Short Momentum

Definition and rules. The Long-Short Momentum strategy takes an active position in both directions: long (+1) when the model predicts "Up" and short (−1) when the model predicts "Down." There is no flat position; the strategy is always in the market.

Signal	Position	Interpretation
Predicted Up ($\hat{y}_t = 1$)	Long +1	Buy wheat futures for one day
Predicted Down ($\hat{y}_t = 0$)	Short −1	Sell wheat futures for one day

Economic intuition. Long-Short strategies are grounded in time-series momentum and cross-sectional momentum literature. Moskowitz et al. [2012] show that the sign of past returns predicts future returns with positive expected returns to both the long and short sides. Asness et al. [2012] demonstrate that risk-parity-style allocation across long-short positions can improve Sharpe ratios by exploiting the full information in the signal. In a daily direction-prediction context, going short on predicted down-days extracts return from both sides of the market, doubling the information ratio of a correct prediction relative to Long-Only, but also doubling the cost of an incorrect one.

Signal model: VAR. The VAR signal maps directly: $\hat{y}_t = 1 \Rightarrow +1$; $\hat{y}_t = 0 \Rightarrow -1$. With the corrected VAR's test accuracy of 47.37%, the expected return per trade is in fact *negative*:

$$\mathbb{E}[\text{return}] = 0.4737 \times r^+ - 0.5263 \times r^- < 0$$

where r^+ and r^- are the average absolute returns on up and down days respectively.

This is consistent with the observed negative Long-Short returns and reinforces that the corrected VAR does not generate a tradeable edge in this period.

Signal model: Transformer. The Transformer's near-constant "Down" prediction translates into a near- constant short position in the Long-Short strategy. During the test period (which included wheat price movements in both directions), a persistent short would generate profits when prices fall and losses when prices rise. The strategy thus becomes a stress test of the cost of class collapse: rather than missing up-days (as in Long-Only), the Transformer Long-Short actively loses money on them.

Advantages of Long-Short Momentum.

- **Full market participation**: profits from both up and down moves, not just up-moves as in Long-Only.
- **Signal stress test**: model bias is immediately visible as persistent positioning and its P&L consequences, making it a useful diagnostic tool.
- **Higher Sharpe potential**: correctly exploiting both sides improves the information ratio relative to Long-Only, since more trading days contribute positive expected return.

Limitations of Long-Short Momentum.

- **Amplified wrong-signal cost**: a false short loses from both the short leg *and* the missed long opportunity — effectively a 2× impact relative to Long-Only.
- **Short-selling costs**: margin requirements, roll costs, and borrowing fees for short wheat futures positions are not modelled here and would erode already thin returns.
- **Near-random signal amplification**: with the corrected VAR accuracy below 50% on the test set, the Long-Short strategy amplifies losses rather than gains; any accuracy below 50% produces negative expected returns on both legs.
- **Transformer class collapse**: the Transformer's persistent short generates large drawdowns on any up-trending period.

Performance metrics.

Table 3: Long-Short Strategy — backtest results, test period (3 Jan – 28 Nov 2025, 228 days).

Metric	**VAR Long-Short**	**Transformer Long-Short**	**Buy & Hold**
Cumulative Return	−8.55%	−10.46%	−4.26%
Annual Return	−9.45%	−11.54%	−4.72%
Sharpe Ratio	−0.293	−0.390	−0.082
Max Drawdown	−17.19%	−17.16%	−19.49%
Num Trades	124	5	1
Win Rate	50.44%	50.00%	46.02%

Comparison with buy-and-hold. Both strategies should be benchmarked against a passive buy-and-hold position, which simply holds one unit of ZW=F throughout the test period. The buy-and-hold captures the full wheat futures return with no transaction costs and no model risk. For either active strategy to be economically justified, it must exceed the buy-and-hold return on a risk-adjusted basis (i.e., higher Sharpe ratio) or offer materially lower drawdown.

An important reflection: *higher predictive accuracy does not necessarily produce better trading performance.* A model with 55% accuracy but large errors on high-volatility days may underperform a 52% accurate model whose errors cluster on low-return days. The relationship between classification accuracy and economic value depends on the interaction between the signal and the return distribution — a nuance that motivates reporting both statistical and economic metrics.

5 Evaluation and Results Summary

5.1 Statistical Forecasting Performance

Table 4 provides a consolidated view of the statistical evaluation metrics across all four models. These metrics characterise the quality of the probabilistic forecast independently of any trading rule.

Table 4: Consolidated statistical evaluation metrics, test set (228 days). "CC" = class collapse.

Model	**Accuracy**	**Precision^{Up}**	**Recall^{Up}**	**F1^{Up}**	**ROC-AUC**	**Note**
Decision Tree	≈ 0.50	≈ 0.50	≈ 0.50	≈ 0.50	≈ 0.50	Baseline
VAR (midterm)	0.5307	0.4923	0.6095	0.5447	0.5438	Arbitrary vars
VAR (final)	0.4737	—	—	—	0.5283	Best AUC
GRU	≈ 0.50	low	≈ 0	≈ 0	≈ 0.50	CC
Transformer	0.5263	low	0.18	—	0.4692	CC; AUC < 0.5

Metric interpretations.

- **Accuracy**: proportion of correct direction predictions. Near 50% is expected in efficient markets.
- **Precision**: of all "Up" predictions, what fraction were correct. Low precision for the minority class in GRU/Transformer reflects the rarity of non-Down predictions.
- **Recall**: of all true "Up" days, what fraction were predicted correctly. Near-zero recall confirms class collapse for GRU and Transformer.
- **F1-score**: harmonic mean of precision and recall, penalising models that achieve one at the expense of the other.
- **ROC-AUC**: area under the receiver operating characteristic curve; 0.5 is random, 1.0 is perfect. The corrected VAR's AUC of 0.5283 indicates a small but genuine lift above random. Notably, the Transformer's AUC of 0.4692 is *below* random, confirming its Down-biased predictions are negatively correlated with true direction.

5.2 Statistical vs. Economic Performance

The distinction between statistical forecasting quality and economic value is central to the evaluation. Statistical metrics assess whether the model's predictions correlate with realised outcomes. Economic metrics assess whether exploiting those predictions generates superior risk-adjusted returns.

Even VAR's modest statistical edge does not guarantee economic value: transaction costs, financing costs for leverage, and the specific distribution of returns on correctly vs. incorrectly predicted days all affect the P&L outcome. Conversely, a model with only 51% accuracy could generate significant economic value if its correct predictions consistently occur on high-return days.

This project highlights one channel through which statistical accuracy fails to translate into trading performance: class collapse in the Transformer causes the Long-Only strategy to remain almost entirely out of the market (4 trades), generating a -5.54% return while

missing nearly all up-days. The disconnect is not due to model accuracy per se but to the specific distribution of signals — a reminder that the evaluation of investment strategies requires economic, not just statistical, lenses.

6 Conclusion

6.1 Main Findings

This project set out to forecast the daily direction of U.S. Wheat Futures prices using four machine learning models and to translate those forecasts into two investment strategies. The key findings are:

1. **VAR outperforms deep learning models** on this dataset. With 3,502 training observations, VAR's 80-parameter specification is far better identified than the GRU and Transformer, which have thousands of parameters relative to only 161–185 effective rolling-window sequences. Model parsimony dominates representational capacity when training data is scarce.
2. **Class collapse is a critical failure mode** for the GRU and Transformer. Both models predict "Down" almost exclusively on the test set due to class imbalance in the binary direction target combined with unweighted binary cross-entropy loss. The primary fix — adding `pos_weight` in PyTorch's `BCEWithLogitsLoss` — is straightforward but was not applied in the submitted versions.
3. **Near-random accuracy is expected and defensible**. Daily price direction in liquid futures markets is inherently close to a 50/50 outcome, consistent with the semi-strong form of the Efficient Market Hypothesis. The Decision Tree's ≈50% accuracy serves as a valid baseline. Importantly, the Transformer's 52.63% accuracy is also misleading — it is driven entirely by Down-bias on a Down-majority test set, as confirmed by its sub-random AUC of 0.4692.
4. **Variable selection critically determines VAR performance**. The midterm VAR using the first 5 FRED-MD columns by position (RPI, W875RX1, CMRMT-SPLx, IPFPNSS, USWTRADE) achieved 53.07% accuracy and AUC 0.5438. After correcting to economically motivated variables (FEDFUNDS, GS10, BAA, TB3MS), accuracy fell to 47.37% but AUC remained above random at 0.5283 — a more honest and defensible result. The higher figure from the midterm version was partly spurious.
5. **Predictive accuracy and trading performance are not equivalent**. The corrected VAR's test accuracy of 47.37% produces a negative expected return in the Long-Short strategy, consistent with the observed −8.55% cumulative return. The Transformer's 4-trade Long-Only and 5-trade Long-Short strategies confirm that class collapse renders a model economically useless regardless of its headline

accuracy figure.

6.2 Limitations

- **Transaction costs not modelled**: futures bid-ask spreads, brokerage commissions, and roll costs would erode or eliminate the thin VAR edge in practice.
- **No position sizing**: binary signals (flat or full) are suboptimal; Kelly criterion or volatility-scaled sizing would better manage the risk-return trade-off.
- **GRU and Transformer not fixed**: the class weighting fix was identified but not applied before submission; corrected versions would provide a fairer comparison of deep learning potential.
- **FRED-MD variable constraints**: theoretically well-motivated variables (INDPRO, CPIAUCSL, UNRATE) are not in the 31-feature selection, limiting the economic grounding of the VAR endogenous system.
- **Limited test period**: the test set covers only 3 January – 28 November 2025 (228 trading days), which coincided with a bear market for wheat futures (buy-and-hold: -4.26%) and may not be representative of longer-horizon or bull-market performance.

6.3 Future Directions

- Apply `pos_weight` class weighting to GRU and Transformer losses and re-evaluate to obtain a fair deep learning baseline.
- Extend the feature set with weather data, supply-chain indicators, or satellite-derived crop condition indices — alternative datasets with direct physical linkage to wheat supply shocks.
- Implement volatility-scaled position sizing (as in Asness et al. 2012) to normalise risk across high- and low-volatility regimes.
- Incorporate transaction cost models and test whether any strategy survives realistic cost assumptions.
- Explore ensemble approaches that combine VAR directional signals with GRU/Transformer probability estimates, potentially capturing complementary information.

References

Asness, C., Frazzini, A., & Pedersen, L. H. (2012). Leverage aversion and risk parity. *Financial Analysts Journal*, 68(1), 47–59.

Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.

McCracken, M. W., & Ng, S. (2016). FRED-MD: A monthly database for macroeconomic research. *Journal of Business & Economic Statistics*, 34(4), 574–589.

Moskowitz, T. J., Ooi, Y. H., & Pedersen, L. H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2), 228–250.

Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance*, 19(3), 425–442.

Sims, C. A. (1980). Macroeconomics and reality. *Econometrica*, 48(1), 1–48.

Cho, K., van Merrienboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using RNN encoder-decoder for statistical machine translation. *arXiv preprint arXiv:1406.1078*.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.